

PROJECT REPORT

Real-Time Attendance System

CSE1021

POONKUNTRAN S
KISHORE KUMAR V

Real-Time Attendance System - Code Summary

Project Overview

A simple Python console-based application that records student or employee attendance with live updates. Users can mark attendance as "Present" or "Absent" and see real-time updates of all recorded attendance.

Code Breakdown

Line 1: Initialize Dictionary

```
attendance_record = {}
```

- Creates an empty dictionary to store attendance records.
 - **Format:** {name: status} (e.g., {"John": "Present", "Alice": "Absent"})
-

Line 3: Define Function

```
def mark_attendance():
```

- Defines a function named `mark_attendance()` that handles the entire attendance marking process.
-

Line 4: Welcome Message

```
print("Enter attendance (type 'exit' to quit):")
```

- Displays instructions to the user.
-

Line 5: Main Loop

```
while True:
```

- Creates an infinite loop that continues until the user types "exit".
-

Line 6-7: Get Name Input

```
name = input("Enter name: ").strip()
if name.lower() == 'exit':
```

- Prompts user to enter a name.
 - `.strip()` removes extra spaces.
 - Checks if user typed "exit" (case-insensitive) to stop the program.
-

Line 8-9: Exit Program

```
print("Exiting attendance system...")
break
```

- If user enters "exit", prints goodbye message and exits the loop.
-

Line 11-12: Get Attendance Status

```
status = input("Enter status (Present/Absent): ").strip().capitalize()
if status not in ['Present', 'Absent']:
```

- Prompts user to enter attendance status.
 - `.capitalize()` converts input to proper format (e.g., "present" becomes "Present").
 - Checks if input is valid (either "Present" or "Absent").
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Line 13-14: Validate Input

```
print("Invalid status! Please enter 'Present' or 'Absent'.")
continue
```

- If status is invalid, shows error message.
 - `continue` skips to the next iteration of the loop without saving invalid data.
-

Line 16: Save Attendance

```
attendance_record[name] = status
```

- Saves the name and status to the dictionary.
 - If name already exists, it updates the status.
-

Line 17-20: Display Current Attendance

```
print("\nCurrent Attendance:")
for person, stat in attendance_record.items():
    print(f"{person}: {stat}")
print()
```

- Displays all recorded attendance in real-time.
 - `for` loop iterates through each entry in the dictionary.
 - `{person}: {stat}` shows name and status.
 - Empty `print()` adds spacing for readability.
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Line 22: Run the Program

```
mark_attendance()
```

- Calls the `mark_attendance()` function to start the program.
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How It Works - Step by Step

1. **Start:** Program initializes empty attendance dictionary.
 2. **Input Name:** User enters a person's name.
 3. **Check Exit:** If name is "exit", program stops.
 4. **Input Status:** User enters attendance status (Present/Absent).
 5. **Validate:** If status is invalid, user must re-enter.
 6. **Save:** Attendance is saved to dictionary.
 7. **Display:** All current attendance records are shown.
 8. **Repeat:** Loop continues for next entry.
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Code

```
attendance_record = {}

def mark_attendance():
    print("Enter attendance (type 'exit' to quit):")
    while True:
        name = input("Enter name: ").strip()
        if name.lower() == 'exit':
            print("Exiting attendance system...")
            break

        status = input("Enter status (Present/Absent): ").strip().capitalize()
        if status not in ['Present', 'Absent']:
            print("Invalid status! Please enter 'Present' or 'Absent'.")
            continue

        attendance_record[name] = status
        print("\nCurrent Attendance:")
        for person, stat in attendance_record.items():
            print(f"{person}: {stat}")
        print()

mark_attendance()
```

Sample Run

```
Enter attendance (type 'exit' to quit):  
Enter name: John  
Enter status (Present/Absent): Present  
  
Current Attendance:  
John: Present  
  
Enter name: Alice  
Enter status (Present/Absent): Absent  
  
Current Attendance:  
John: Present  
Alice: Absent  
  
Enter name: Bob  
Enter status (Present/Absent): invalid  
Invalid status! Please enter 'Present' or 'Absent'.  
Enter status (Present/Absent): Present  
  
Current Attendance:  
John: Present  
Alice: Absent  
Bob: Present  
  
Enter name: exit  
Exiting attendance system...
```

Key Features

- ✓ **Simple Console Interface:** Easy to use without web technologies.
 - ✓ **Real-Time Updates:** Shows attendance instantly after each entry.
 - ✓ **Input Validation:** Prevents invalid status entries.
 - ✓ **Case-Insensitive:** Accepts "exit", "EXIT", or "Exit".
 - ✓ **No External Dependencies:** Uses only Python built-in functions.
 - ✓ **Data Storage:** Keeps attendance in memory during runtime.
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Skills Demonstrated

- Dictionary data structure
 - While loops and control flow
 - User input handling
 - String manipulation (.strip(), .capitalize(), .lower())
 - Conditional statements (if, elif, continue, break)
 - For loops and iteration
 - String formatting (f-strings)
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Future Enhancements

- Save attendance to a CSV or text file.
 - Generate attendance reports.
 - Search for specific person's attendance.
 - Calculate present/absent percentages.
 - Use a database instead of dictionary.
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Conclusion

This is a beginner-friendly real-time attendance system that demonstrates core Python programming concepts.

References

- Vityarthi
- Official python
- GeeksforGeeks python tutorials
- google